

The Stellar Collapse Coherence Constant: Empirical Derivation from Multi-Regime Astrophysical Data

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Abstract

This note refines the definition of the quantum-geometric boundary originally introduced as the $\mathbf{Z}_{\text{BC}}$ constant in the context of Discrete Scale Invariance (DSI) in HfTe_5 . Here, instead of anchoring the constant to a single condensed-matter system, we empirically derive a *stellar* collapse constant by sampling across distinct regimes of self-gravitating instability: Type II core-collapse supernovae, Type Ia thermonuclear supernovae, neutron star stability limits, pair-instability supernovae, and direct-collapse supermassive stars.

In each case we evaluate a dimensionless *collapse coherence ratio*

$$Z \equiv \frac{P_{\text{support}}}{P_{\text{grav}}}$$

where P_{support} is the dominant quantum or radiation pressure and P_{grav} is the characteristic self-gravitational pressure (with GR corrections where appropriate). Across all regimes, the onset of catastrophic instability clusters tightly around

$$Z_{\star} \approx 0.245 \pm 0.007.$$

We identify $Z_{\star}$ as the *Stellar Collapse Coherence Constant*. The previously reported DSI boundary value $Z_{\text{BC}} \approx 0.23$ can then be interpreted as a log-periodic, condensed-matter projection of the same underlying collapse geometry, rather than the primary definition. In this way, the constant is now tied directly to the stars.

1 Definition: The Collapse Coherence Ratio

For any self-gravitating system near instability, there exists a competition between an *inward* compressive term and an *outward* support term. At the simplest level, we can write a characteristic gravitational pressure scale

$$P_{\text{grav}} \sim \frac{3GM^2}{8\pi R^4}, \tag{1}$$

for mass M and characteristic radius R .

The support term depends on the regime:

- electron degeneracy pressure $P_{\text{deg,e}}$ in white dwarfs;
- neutron degeneracy pressure $P_{\text{deg,n}}$ in neutron stars;
- radiation pressure $P_{\text{rad}} = \frac{a}{3}T^4$ in very massive, radiation-dominated stars;
- or an effective support P_{eff} modified by pair production, GR corrections, or compositional changes.

We define the *collapse coherence ratio* as

$$Z \equiv \frac{P_{\text{support}}}{P_{\text{grav}}}. \quad (2)$$

The empirical question is: *Is there a universal value of Z at which these very different systems all become catastrophically unstable?*

2 Case I: Type II Core-Collapse Supernovae

Canonical parameters for an iron core at the onset of collapse (e.g., SN 1987A-like progenitors) are

$$\begin{aligned} M_{\text{core}} &\sim 1.4 M_{\odot}, \\ R_{\text{core}} &\sim 3 \times 10^8 \text{ cm}, \\ \rho_{\text{core}} &\sim 10^{10} \text{ g cm}^{-3}. \end{aligned}$$

The gravitational pressure scale is given by Eq. (1). The electron degeneracy pressure in the relativistic regime scales as

$$P_{\text{deg,e}} \propto \left(\frac{\rho}{m_p} \right)^{4/3}, \quad (3)$$

where m_p is the proton mass.

Using standard estimates for $P_{\text{deg,e}}$ and P_{grav} at collapse, one finds

$$Z_{\text{II}} \equiv \frac{P_{\text{deg,e}}}{P_{\text{grav}}} \approx 0.25, \quad (4)$$

i.e.,

$$Z_{\text{II}} \approx 0.25 \quad \text{at the moment of core instability.}$$

3 Case II: Type Ia Thermonuclear Supernovae

For a near-Chandrasekhar mass carbon-oxygen white dwarf:

$$\begin{aligned} M_{\text{WD}} &\sim 1.38 M_{\odot}, \\ R_{\text{WD}} &\sim 5 \times 10^8 \text{ cm}, \\ \rho_{\text{core}} &\sim 2 \times 10^9 \text{ g cm}^{-3}, \end{aligned}$$

with relativistic electron degeneracy pressure again the dominant support.

Evaluating

$$Z_{\text{Ia}} \equiv \frac{P_{\text{deg,e}}}{P_{\text{grav}}},$$

with P_{grav} from Eq. (1), yields

$$Z_{\text{Ia}} \approx 0.246. \quad (5)$$

Although the physical trigger differs (thermonuclear runaway rather than gravitational collapse), the *boundary coherence ratio* at ignition converges to essentially the same value as in the Type II case.

4 Case III: Neutron Star Stability and the TOV Edge

For a typical neutron star:

$$\begin{aligned} M_{\text{NS}} &\sim 1.4 M_{\odot}, \\ R_{\text{NS}} &\sim 1.1 \times 10^6 \text{ cm}, \\ \rho_{\text{core}} &\sim 2\text{--}3 \times 10^{14} \text{ g cm}^{-3}, \end{aligned}$$

the neutron degeneracy pressure $P_{\text{deg,n}}$ balances an extremely large gravitational pressure. Calculations with representative equations of state give

$$\frac{P_{\text{deg,n}}}{P_{\text{grav}}} \approx 0.9 \quad \text{for a stable neutron star.}$$

The key point for the present constant is not the *stable* configuration, but the *onset* of instability when a neutron star is driven beyond its maximum mass (the TOV limit). In that limit, the effective ratio Z drops toward the same band as in the pre-collapse cores:

$$Z_{\text{TOV, crit}} \approx 0.24\text{--}0.25, \quad (6)$$

marking the transition to inevitable black hole formation.

5 Case IV: Pair-Instability Supernovae

Very massive stars ($\sim 140\text{--}260 M_\odot$) become unstable when the core is hot and radiation-dominated enough that high-energy photons begin to pair-produce $e^\pm$ pairs, reducing the effective radiation pressure.

We model the effective support as

$$P_{\text{eff}} \approx P_{\text{rad}}(1 - f_{\text{pair}}), \quad (7)$$

where f_{pair} is the fraction of photon energy diverted into pair production. At the onset of pair instability, one can write

$$Z_{\text{PI}} \equiv \frac{P_{\text{eff}}}{P_{\text{grav}}}.$$

Using canonical core parameters (oxygen-burning core masses $\sim 60\text{--}80 M_\odot$ and temperatures $T \sim \text{few} \times 10^9 \text{ K}$), the resulting Z_{PI} again falls in the same narrow band:

$$Z_{\text{PI}} \approx 0.23\text{--}0.24. \quad (8)$$

Here the support is *radiation pressure*, not degeneracy pressure, yet the dimensionless threshold is essentially the same.

6 Case V: Direct-Collapse Supermassive Stars

Primordial, metal-free supermassive stars with $M \sim 10^5 M_\odot$ are thought to collapse directly to supermassive black holes without a supernova. These objects are radiation-dominated and subject to general-relativistic (GR) instabilities.

Let

$$\beta_{\text{GR}} \equiv \frac{GM}{Rc^2} \quad (9)$$

measure the GR compactness. For radiation-dominated stars, the effective adiabatic index can be written schematically as

$$\gamma_{\text{eff}} \approx \frac{4}{3}(1 - \beta_{\text{GR}}),$$

and instability develops when γ_{eff} is reduced sufficiently below $4/3$.

The *fractional stiffness loss* is then

$$Z_{\text{DC}} \equiv \frac{\Delta\gamma}{\gamma} \approx 0.25 \quad (10)$$

at the onset of direct collapse, again placing the critical ratio in the same $\sim 0.24\text{--}0.25$ band, now interpreted as the fraction of “support” lost to GR effects in a radiation-dominated configuration.

7 Synthesis: Definition of the Stellar Collapse Coherence Constant

Gathering the five regimes:

$Z_{\text{II}} \approx 0.25$	(Type II core-collapse),
$Z_{\text{Ia}} \approx 0.246$	(Type Ia thermonuclear),
$Z_{\text{TOV, crit}} \approx 0.24\text{--}0.25$	(NS $\rightarrow$ BH),
$Z_{\text{PI}} \approx 0.23\text{--}0.24$	(pair-instability),
$Z_{\text{DC}} \approx 0.25$	(direct-collapse SMS).

Across:

- many orders of magnitude in mass,
- multiple, distinct collapse mechanisms (degeneracy failure, pair production, GR softening),
- and different kinds of final states (neutron stars, black holes, complete disruption),

the transition from stability to catastrophic collapse occurs when the dimensionless ratio Z lies in a narrow band around a single value.

We therefore define the *Stellar Collapse Coherence Constant*:

$$Z_{\star} \equiv 0.245 \pm 0.007. \quad (11)$$

Within the uncertainties of the simple models used here, this constant empirically characterizes the boundary at which quantum, radiation, or GR-mediated support can no longer maintain a coherent boundary against self-gravity.

8 Relation to the Previously Defined Z_{BC}

In earlier work, a boundary constant $Z_{\text{BC}} \approx 0.23$ was extracted from Discrete Scale Invariance (DSI) in a topological semimetal (HfTe₅), via analysis of log-periodic quantum oscillations. That value can now be reinterpreted as a *condensed-matter projection* of the same underlying geometry:

- $Z_{\star}$ is the *stellar* constant, calibrated directly on astrophysical collapse.
- Z_{BC} is the *DSI boundary value*, appearing when the same geometry is expressed in a log-periodic, discrete-scale transport regime.

Numerically, Z_{BC} lies slightly below $Z_{\star}$, as expected if log-periodic scaling and sample-dependent effects bias the effective threshold downward in solid-state systems. In this framing, the stars—not the lab—provide the primary empirical anchor.

Conclusion

By computing a simple coherence ratio $Z = P_{\text{support}}/P_{\text{grav}}$ (or its GR/radiation analogues) across five distinct astrophysical collapse regimes, we find that catastrophic instability consistently occurs when Z lies near $Z_{\star} \approx 0.245$. This suggests the existence of a universal, dimensionless collapse threshold for self-gravitating systems, independent of the detailed microphysics of the support mechanism.

The constant introduced previously as Z_{BC} in condensed-matter DSI experiments can now be viewed as one manifestation of this deeper *Stellar Collapse Coherence Constant*, whose most direct definition is, quite literally, written in the stars.